

Question			Marking details	Marks available					
				AO1	AO2	AO3	Total	Maths	Prac
3	(a)	(i)	Horizontal – constant speed (with or without reference to force) (1) Vertical – [constant] acceleration due to force of gravity [acting downwards] (1)	2			2		
		(ii)	Similarity- projectile will [still] hit the apple (1) Difference - the apple falls further before being hit or greater time [before being hit] (1) Award 2 marks for - The apple will be hit further down		2		2		
	(b)	(i)	$v_H = 3.0 \text{ [m s}^{-1}\text{]}$	1			1	1	
		(ii)	$v_v = 2.1 \text{ [m s}^{-1}\text{]}$ (from $3\tan 35^\circ$) (ecf on v_H) other methods possible (1) Attempt at use of $v^2 = u^2 + 2ax$ including allowing for incorrect use of signs (1) Correct substitution e.g. $0 = (2.1)^2 - 19.62h$ (1) (ecf on u_v) $h = 0.22 \text{ [m]}$ (1) Alternative: $v_v = 2.1 \text{ [m s}^{-1}\text{]}$ (from $3\tan 35^\circ$) (ecf on v_H) other methods possible (1) Attempt at use of $x = ut + \frac{1}{2}at^2$ including allowing for incorrect use of signs (1) Correct substitution: $h = 2.1 \times 0.2 - 4.9 \times 0.2^2$ (ecf on u_v) (1) $h = 0.224 \text{ [m]}$ (1)	1 1	1 1		4	4	

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			Alternative: $v_v = 2.1 \text{ [m s}^{-1}\text{]}$ (from $3\tan 35^\circ$) (ecf on v_H) other methods possible (1) Attempt at use of $h = \frac{(u+v)}{2}t$ including allowing for incorrect use of signs (1) Correct substitution: $h = \frac{(2.1 \times 0.2)}{2}$ (ecf on u_v) (1) $h = 0.21 \text{ [m]}$ (1) Alternative: $v_v = 1.962 \text{ [m s}^{-1}\text{]}$ using $v = u + at$ (1) Attempt at use of $h = \frac{(u+v)}{2}t$ (1) Correct substitution: $h = \frac{(1.962 \times 0.2)}{2}$ (ecf on u_v) (1) $h = 0.196 \text{ [m]}$ (1) Alternative: Attempt at use of $x = ut + \frac{1}{2}at^2$ including allowing for incorrect use of signs (1) $u = 0$ (or $ut = 0$) (1) Correct substitution: $h = 0 + \frac{1}{2} \times 9.81 \times 0.2^2$ (1) $h = 0.196 \text{ [m]}$ (1)						

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		(iii)	Concept: $E_{\text{total}} = E_k + E_p$ (1) Subs: $E_{\text{total}} = \frac{1}{2} \times 0.1 \times (3)^2 + 0.1 \times 9.81 \times 0.22$ (1) ecf on v_H but not incorrect v used, and h . $E_{\text{total}} = 0.67$ [J] (1) Alternative Total E at h = initial KE (1) can be implied $= \frac{1}{2} \times 0.1 \times 3.66^2$ (1) $= 0.67$ [J] (1)	1	1 1		3	3	
	(c)		Energy at C = $0.5 \times 0.1 \times (3.3)^2$ (1) $= 0.54$ [J] (1) Valid method for showing loss is approximately 20% e.g. 0.67 ecf $\times 0.8 = 0.54$ [J] so claim correct (1) Alternative: Ratio of KE attempted (1) Answer = 0.81 or 81% (1) Hence 19% drop so claim correct (1) Alternative: $0.8 \times 0.67 = 0.536$ [J] (1) $v = \sqrt{\frac{2 \times 0.536}{0.1}}$ (1) $v = 3.27$ [m s ⁻¹] close to 3.3 so claim correct (1)			3	3	3	
			Question 3 total	6	6	3	15	11	0